

# Calibrating the Decision, Not the Model

Per-class thresholds and argmax scaling for training-free  
open-vocabulary segmentation of remote sensing imagery

Priyanshu Kumar   Raj Gautam   Shounak Chakraborty   K Sathya Babu

Indian Institute of Information Technology Design and Manufacturing, Kurnool

September 9, 2026

Introduction

Motivation & Applications

Problem Statement

Literature Survey

Research Gaps

Objectives

Experimental Setup & Results

References

# 1. Introduction — the task

## Open-vocabulary semantic segmentation of aerial imagery.

- Input: an image *and a list of class names, as plain words*.
- Output: a class for every pixel.
- **Training-free / annotation-free**: no weights are trained. The class list is supplied at inference and can change without retraining.

**Why it matters.** A model trained for  $N$  fixed classes cannot answer a question outside them. Land cover mapping constantly needs new classes — solar panel, flood water, debris — and labelled data for them rarely exists. **The pipeline we build on.** SAM 3 emits, per class

prompt: a presence score  $S_{\text{pres}}$ , a dense semantic map, and instance masks. SegEarth-OV3 fuses these and thresholds the result.

## 1. Introduction — the contribution in one sentence

A single confidence threshold is the wrong *shape* for a multi-class open-vocabulary pipeline. Fitting **one threshold per class**, and **one scale per class before the argmax**, on the same supervision budget the baseline already spends tuning its single  $\tau$  — **with no weights trained** — recovers a real gain.

|                           | LoveDA        | Potsdam      | OpenEarthMap |
|---------------------------|---------------|--------------|--------------|
| SegEarth-OV3 (reproduced) | 47.38         | 57.60        | 44.16        |
| + per-class $\tau$        | + 1.18        | 58.35        | +0.16        |
| + argmax scaling          | + <b>2.32</b> | <b>63.27</b> | —            |

All verified end-to-end by the official evaluator on held-out tiles, not by cache arithmetic.

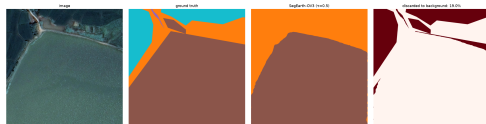
## 2. Motivation — the baseline throws land cover away

On LoveDA validation, at the published  $\tau = 0.5$ :

- **29.68%** of all real-class pixels are assigned to background — 323 million pixels.
- Discard outnumbers real-class confusion **3:1**. The dominant error is *silence*, not error.
- water: precision **89.5%**, recall **54.7%**. Right nine times in ten when it fires; finds half of what is there.

Every global fix costs more than it returns:

- $\tau \rightarrow 0.1$ : **-5.54 mIoU**, 1.73 wrong per 1 right
- remove presence gating: **-11.97 mIoU**



LoveDA tile: image / ground truth / baseline / discarded pixels. Whole water regions dropped to background.

## 2. Applications — why a cheap correction matters

**Disaster response.** Flood extent from UAV imagery within hours. No labelled data for the affected district; no time to train.

**Renewable-energy infrastructure.** solar panel is a class no land-cover benchmark contains. Supplied as two words.

**Informal settlement mapping.** tin roof, unpaved road, water tank — vocabulary that changes per survey.

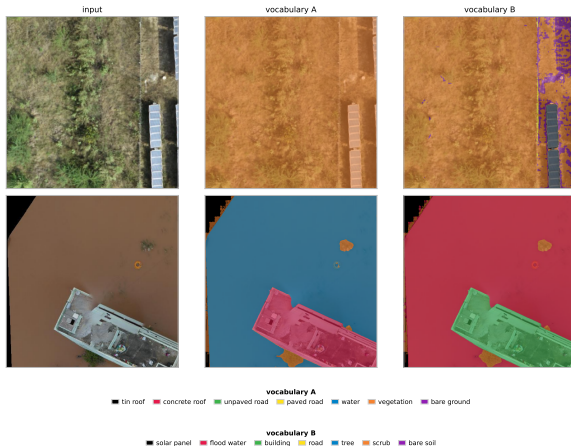
**Cost of our correction:** ~200 labelled tiles, minutes of CPU, **no gradient steps**. Cost of fine-tuning: thousands of tiles, a training run, and the training-free property is lost.

Measured, on Indian UAV imagery

25 tiles, 5 openly licensed scenes, 4–5 cm — a region absent from all three benchmarks. Run unchanged under two class lists typed at inference.

- **The partition is stable under renaming:** 78.6% water vs 79.1% flood water; 18.2% concrete roof vs 18.3% building.
- **The panels are only partly found:** 100.0% of the solar-farm tile goes to one class when no word for them exists; 4.6% with solar panel supplied.

## 2. Applications — one scene, two vocabularies



Same weights, same forward pass; only the class list differs. **Bottom:** a flooded building — both vocabularies recover the same regions and only the names follow the words. **Top:** a photovoltaic array the settlement vocabulary has no word for. **No ground truth exists for this imagery**, so these are pixel shares, not accuracy.

### 3. Problem statement — formulation

Let  $\mathcal{C} = \{1, \dots, N\}$  be the vocabulary and  $s_c(p) \in [0, 1]$  the score the pipeline assigns class  $c$  at pixel  $p$ .

**The baseline decision rule** (one global  $\tau$ ):

$$\hat{y}(p) = \begin{cases} \arg \max_c s_c(p), & \text{if } \max_c s_c(p) \geq \tau \\ b \text{ (catch-all),} & \text{otherwise} \end{cases}$$

**What we fit instead**, on a calibration set of  $\sim 200$  labelled tiles:

$$\hat{y}(p) = \begin{cases} c^*, & \text{if } s_{c^*}(p) \geq \tau_{c^*} \\ b, & \text{otherwise} \end{cases} \quad c^* = \arg \max_c \mathbf{w}_c s_c(p)$$

#### Objective

$(\mathbf{w}, \tau) = \arg \max \text{mIoU}_{\text{real}}$  by coordinate ascent. **No network weights are touched.** The threshold reads the *raw* score  $s_{c^*}$ , so  $\mathbf{w}$  acts only on the  $\arg \max$ .



### 3. Why per-class $\tau$ is the *complete* family — and $w$ is not

**Claim.** Given a fixed  $\text{argmax}$ , any strictly increasing per-class recalibration  $g_c$  composed with a global  $\tau$  collapses to a per-class threshold vector:

$$g_c(s) \geq \tau \iff s \geq g_c^{-1}(\tau) = \tau_c.$$

- So **per-class  $\tau$  exhausts everything reachable by rescaling scores after the  $\text{argmax}$ .** Nothing in that family can beat it.
- Scaling *before* the  $\text{argmax}$  changes *which* class wins, so it is a **strictly larger family** — it reaches the errors a threshold provably cannot.

#### Measured consequence (LoveDA, §7.7)

6.0% of catch-all assignments are cases where the catch-all *won the  $\text{argmax}$*  at  $s \geq \tau$ .

**Lowering a threshold cannot change an  $\text{argmax}$ ,** so that residual is unreachable by lever one and is exactly what lever two addresses.

## 4. Literature survey — the lineage (1/2)

| Work          | Venue     | Contribution / relevance                                                         |
|---------------|-----------|----------------------------------------------------------------------------------|
| CLIP [1]      | ICML 2021 | Image-text contrastive pretraining; the origin of the open-vocabulary paradigm.  |
| SAM [2]       | ICCV 2023 | Promptable class-agnostic segmentation; masks without a fixed label set.         |
| SAM 3 [3]     | 2025      | Presence + semantic + instance heads in one model. <a href="#">Our backbone.</a> |
| MaskCLIP [4]  | ECCV 2022 | First to extract dense predictions from CLIP by removing pooling.                |
| SCLIP [5]     | 2023      | Correlative self-attention; the codebase SegEarth-OV3 derives from.              |
| ClearCLIP [6] | ECCV 2024 | Isolates which CLIP components harm dense prediction.                            |
| ProxyCLIP [7] | ECCV 2024 | Uses self-supervised features (DINO) as structural guidance.                     |
| GroupViT [8]  | CVPR 2022 | Learns segment grouping from text supervision alone.                             |

## 4. Literature survey — direct competitors and tools (2/2)

| Work                              | Venue                | Contribution / relevance                                                                                               |
|-----------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------|
| <a href="#">SegEarth-OV</a> [9]   | CVPR 2025            | Feature upsampler + global bias subtraction for RS. Defines the protocol.                                              |
| <a href="#">SegEarth-OV3</a> [10] | 2025                 | SAM 3 for RS OVSS: dual-head fusion + presence gating. <b>Our baseline, reproduced at 47.38 vs 47.4.</b>               |
| <a href="#">ConInfer</a> [11]     | 2026                 | GMM over DINOv3 patches fused with the VLM prior; +2.80 mIoU. <b>Nearest competitor — our method composes with it.</b> |
| OVRISBench [12]                   | 2026                 | Unified open-vocabulary RS evaluation protocol.                                                                        |
| DenseCRF [13]                     | NIPS 2011            | The classical “use spatial context” answer we position against.                                                        |
| CRF-as-RNN [14]                   | ICCV 2015            | Learned CRF inference; same family.                                                                                    |
| Calibration [15]                  | ICML 2017            | Modern networks are systematically miscalibrated. <b>The premise we exploit.</b>                                       |
| Otsu [16]                         | IEEE TSMC 1979       | Classical automatic thresholding; one of our label-free baselines.                                                     |
| SLIC [17]                         | IEEE TPAMI 2012      | Superpixels; our region atoms.                                                                                         |
| LoveDA [18] / OEM [19]            | NeurIPS'21 / WACV'23 | The benchmarks.                                                                                                        |

**16 works**; full bibliography (26 entries) in the paper.

## 5. Research gaps

1. **The decision rule is unexamined.** Every method in this family ends in  $\text{argmax} + \text{one global } \tau$ , tuned per dataset. All effort goes into the *scores*; nobody asks whether the *rule* on top of them is the right shape.
2. **The discarded residual is never quantified.** SegEarth-OV3 reports mIoU; it does not report that **29.68%** of real-class pixels on LoveDA are assigned to background.
3. **Catch-all classes distort the metric, in both directions.** mIoU is an *unweighted* mean, so a catch-all owns  $1/N$  of it however meaningless. We measure  $+22.67$  inflating on one dataset and  $-3.51$  deflating on another.
4. **Context is modelled visually, not semantically.** ConInfer uses a per-scene visual prior over patches; nobody tests a corpus-level *semantic* class-adjacency prior. (We did — and it fails. Reported as a negative with an oracle bound.)

## 6. Objectives

1. **Reproduce** SegEarth-OV3 exactly, so every later number is measured against a baseline that runs on our hardware. ✓ 47.38 vs published 47.4
2. **Quantify** the residual the baseline discards, and test whether any global knob reaches it. ✓ 29.68%; all three knobs cost more than they return
3. **Explain** what governs the residual across datasets, by intervention rather than correlation. ✓ vocabulary intervention, arity-controlled, pre-registered
4. **Design** a correction that trains no weights and spends the same supervision budget the baseline already spends. ✓ two levers, +2.32 LoveDA / +5.67 Potsdam
5. **Bound** what is achievable without labels at all. ✓ 12 attempts, all bounded, with a stated mechanism
6. **Demonstrate** on imagery and a vocabulary no benchmark covers. ✓ Indian UAV scenes, 4–5 cm

## 7. Experimental setup

### Datasets

- **LoveDA** — 1669 val tiles,  $1024^2$ , 0.3 m, 7 classes, urban/rural split
- **OpenEarthMap** — 384 tiles, 0.25–0.5 m, 9 classes
- **ISPRS Potsdam** — 2016 tiles, 5 cm, 6 classes
- **OpenAerialMap (India)** — 25 tiles, 4–5 cm, **no ground truth**

### Baselines

- SegEarth-OV3 (SAM 3) — reproduced
- ConInfer (CLIP + DINOv3) — run in a separate environment
- $\tau$  sweep, presence removal, per-class Otsu, 12 label-free rules

### Protocol

- 5-fold cross-validation; calibration and evaluation tiles **always disjoint**
- Baseline recomputed on the *same* held-out tiles
- Both metrics reported: full mIoU *and* catch-all-excluded

### Hardware

- RTX 2000 Ada, 16 GB, 70 W cap
- One LoveDA pass  $\approx$  24 min
- **Fitting the correction: minutes of CPU**

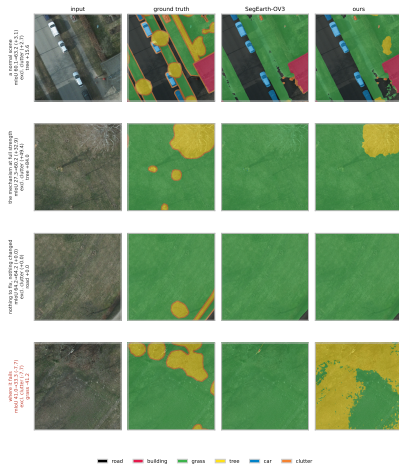
**Validation gate:** every instrumented run must still reproduce 47.37 mIoU and 29.68% discard, or the patch changed behaviour.

## 7. Results — both levers, five-fold

|                           | LoveDA                    | Potsdam                   | OEM                       | ConInfer/LoveDA  |
|---------------------------|---------------------------|---------------------------|---------------------------|------------------|
| Lever 1: per-class $\tau$ | $+1.18 \pm 0.45$          | $+0.59$                   | $+0.30$                   | $+2.51 \pm 0.34$ |
| Lever 2: argmax scaling   | $+1.16 \pm 0.19$          | $+4.86 \pm 0.35$          | —                         | $-0.10$          |
| <b>Total</b>              | <b><math>+2.32</math></b> | <b><math>+5.67</math></b> | $+0.30$                   | $+2.51$          |
| Catch-all-excluded        | $+1.36$                   | $+6.00$                   | <b><math>+1.75</math></b> | $+1.94$          |
| Folds positive            | 5/5                       | 5/5                       | —                         | 5/5              |

- **Two architectures** (SAM 3, CLIP+GMM), **three datasets**, two operating points. **The claim is not about SAM 3.**
- Our method *composes with ConInfer*: their pipeline  $36.99 \rightarrow 39.52$  mIoU.
- **Potsdam tree:  $37.92 \rightarrow 59.09$  IoU** (precision 93.34%, recall 38.63%) — exactly the asymmetry the method targets.

## 7. Results — the method, visually



Four Potsdam tiles. Same weights, same forward pass — only the decision rule differs. **Row four is a tile where the method LOSES 7.72 mIoU** and is included deliberately: the five-fold gain is  $+4.86 \pm 0.35$ , an *average*.



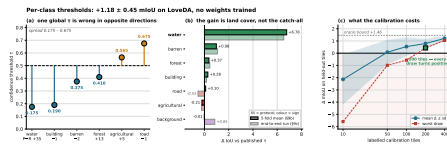
## 7. Results — what cannot be done without labels

### 12 label-free rules tested. All fail.

- Otsu, confidence percentile, presence-scaled, 8 per-class statistics, reachability
- Best: +0.42 against a random control's p95 of +0.58
- Oracle bound: +1.24

### The mechanism, not just the failure:

Precision *is* predictable without labels ( $\rho = +0.943$ ,  $p = 0.017$ ) — and it does not determine the threshold ( $\rho = -0.429$ ,  $p = 0.419$ ). The optimal  $\tau$  solves a **coupled multi-class IoU objective**: raising one class's threshold moves pixels into the catch-all and changes every other class's optimum. **No per-class scalar can express that.**



Fitted thresholds span 0.175–0.675 against a single global 0.5 — one value is wrong for different classes *in opposite directions*.

## 7. Honest limitations

- **Thresholds do not transfer across a domain shift.** Fitted on LoveDA-rural, applied to urban:  $-1.11$  mIoU — *worse than not calibrating*. Pooling both domains also loses most of the gain.
- **The gain is not uniform.** LoveDA rural  $+2.77$ , urban  $+0.10$ . Our headline is a benchmark-level figure, not a claim about every stratum.
- **No label-free rule predicts *whether* calibration will pay** either — so the 200-tile cost buys the answer and the question together.
- **The fitted vectors are constant across tiles**, so they are right on average and can be wrong on any individual scene (Fig. 8, row 4).
- **ConInfer's reproduction is imperfect** — 36.99 against a published 39.33. We report both numbers.

**Every negative in this project is reported with an oracle bound**, so the reader learns where the headroom is rather than only that we failed to reach it.

## 8. References I

-  A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2021.
-  A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, “Segment anything,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023.
-  N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, *et al.*, “SAM 3: Segment anything with concepts,” *arXiv preprint arXiv:2511.16719*, 2025.
-  C. Zhou, C. C. Loy, and B. Dai, “Extract free dense labels from CLIP,” in *Proc. European Conf. Computer Vision (ECCV)*, 2022.
-  F. Wang, J. Mei, and A. Yuille, “SCLIP: Rethinking self-attention for dense vision–language inference,” in *Proc. European Conf. Computer Vision (ECCV)*, 2024.
-  M. Lan, C. Chen, Y. Ke, X. Wang, L. Feng, and W. Zhang, “ClearCLIP: Decomposing CLIP representations for dense vision–language inference,” in *Proc. European Conf. Computer Vision (ECCV)*, 2024.
-  M. Lan, C. Chen, Y. Ke, X. Wang, L. Feng, and W. Zhang, “ProxyCLIP: Proxy attention improves CLIP for open-vocabulary segmentation,” in *Proc. European Conf. Computer Vision (ECCV)*, 2024.
-  J. Xu, S. De Mello, S. Liu, W. Byeon, T. Breuel, J. Kautz, and X. Wang, “GroupViT: Semantic segmentation emerges from text supervision,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022.

## 8. References II

-  K. Li, R. Liu, X. Cao, X. Bai, F. Zhou, D. Meng, and Z. Wang, "SegEarth-OV: Towards training-free open-vocabulary segmentation for remote sensing images," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2025.
-  K. Li, S. Zhang, Y. Wang, Y. Deng, Z. Wang, D. Meng, and X. Cao, "SegEarth-OV3: Exploring SAM 3 for open-vocabulary semantic segmentation in remote sensing images," *arXiv preprint arXiv:2512.08730*, 2025.
-  W. Chen, Z. Hu, Y. Zhang, H. Ning, and Y. Tai, "ConInfer: Context-aware inference for training-free open-vocabulary remote sensing segmentation," *arXiv preprint arXiv:2603.29271*, 2026.
-  B. Li, T. Huo, H. Dong, D. Zhang, Z. Zhao, J. Gao, and X. Li, "Towards realistic open-vocabulary remote sensing segmentation: Benchmark and baseline," *arXiv preprint arXiv:2604.15652*, 2026.
-  P. Krähenbühl and V. Koltun, "Efficient inference in fully connected CRFs with gaussian edge potentials," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2011.
-  S. Zheng, S. Jayasumana, B. Romera-Paredes, V. Vineet, Z. Su, D. Du, C. Huang, and P. H. S. Torr, "Conditional random fields as recurrent neural networks," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2015.
-  C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017.
-  N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

## 8. References III

-  R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 34, no. 11, pp. 2274–2282, 2012.
-  J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.
-  J. Xia, N. Yokoya, B. Adriano, and C. Broni-Bediako, "OpenEarthMap: A benchmark dataset for global high-resolution land cover mapping," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2023.

## What was built

Not a new architecture. **Two lines of arithmetic** on top of an existing pipeline: a per-class threshold vector and a per-class scale before the  $\text{argmax}$ , fitted on  $\sim 200$  labelled tiles with **no gradient steps**.

- +2.32 LoveDA   +5.67 Potsdam   +2.51 on ConInfer's CLIP pipeline
- Verified end-to-end by the official evaluator on held-out tiles
- A proof that per-class  $\tau$  is the complete family given the  $\text{argmax}$
- A causal experiment with pre-registered predictions
- 12 label-free attempts, bounded, with a stated mechanism

**Target venue: IEEE Transactions on Geoscience and Remote Sensing**